Emily J. Williams

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Education

Massachusetts Institute of Technology – Cambridge, MA
PhD, Computational Science and Engineering 2023 – 2026
MS, Aeronautics and Astronautics 2021 – 2023

University of Illinois Urbana-Champaign – Champaign, IL
BS, Aerospace Engineering, with Highest Honors 2017 – 2021
Minor, Atmospheric Sciences 2017 – 2021

Research Experience

Peter O'Donnell Jr. Postdoctoral Fellow – Oden Institute, Austin, TX 2026 – Present
Topics: Stochastic reduced-order modeling using real-world data
Advisor: Karen Willcox

Visiting Researcher – Pacific Northwest National Laboratory, Seattle, WA 2024 – 2026
Topics: Projection-based reduced-order models for multiscale differential equations
Advisors: Panos Stinis and Amanda Howard

DOE Computational Science Graduate Fellow – MIT, Cambridge, MA 2022 – 2026
Topics: Stochastic and generative modeling for multiscale chaotic differential equations
Advisor: David Darmofal

Graduate Research Assistant – MIT, Cambridge, MA 2021 – 2022
Topics: Wall-modeled large-eddy simulation for high-speed flows
Advisor: Adrian Lozano-Duran

Undergraduate Research Assistant – UIUC, Champaign, IL 2019 – 2021
Topics: Numerical methods for fluid-thermal-structure interaction
Advisors: Andres Goza and Marco Panesi

Selected Manuscripts

1. Williams, E., and Darmofal, D., "Generative modeling for closure and linearized stability of chaotic dynamical systems," Preprint, 2025. arXiv:2504.09750.

2. Williams, E., Howard, A., Meuris, B., and Stinis, P., "What do physics-informed DeepONets learn? Understanding and improving training for scientific computing applications," Journal of Computational Physics, 2026. DOI: 10.1016/j.jcp.2026.114851.

3. Williams, E., Arranz, G., Lozano-Duran, A., "Near-Field Wall-Modeled Large-Eddy Simulation of the NASA X-59 Low-Boom Flight Demonstrator," Preprint, 2023. arXiv:2307.02725.

(Full list: https://scholar.google.com/citations?user=6aV6YWgAAAAJhl=en)

Selected Fellowships & Awards

Oden Institute Peter O'Donnell Jr. CES Postdoctoral Fellowship ($160k) 2026 – 2027
Rising Stars in Computational and Data Sciences 2026
PNNL Linus Pauling Distinguished Postdoctoral Fellowship Finalist ($200k) – Declined 2026

DOE Computational Science Graduate Fellowship ($180k)	2022 – 2026
NSF Graduate Research Fellowship ($160k) – Declined	2022
DOD National Defense Science and Engineering Graduate Fellowship ($130k) – Declined	2022
Graduate Student Leadership Award – MIT AeroAstro	2023
AIAA New England Community Award – MIT AeroAstro	2023
Vickie Kerrebrock Award – MIT AeroAstro	2022
Gardner Fellowship – MIT	2022
AIAA Aviation Week Network 20 Twenties	2021
Scott R. White Aerospace Engineering Visionary Scholarship – UIUC	2020
Aerospace Department Academic Scholarship – UIUC	2018
Illinois Engineering Achievement Scholarship – UIUC	2017

Presentations

1. Williams, E., and Darmofal, D., "Generative modeling for closure and linearized stability of chaotic dynamical systems," APS Division of Fluid Dynamics, Interact Session, Nov 2025.

2. Williams, E., Howard, A., and Stinis, P., "Machine-Learning-Based Spectral Methods Toward Reduced-Order Modeling for Partial Differential Equations," SIAM Computational Science & Engineering, Mar 2025.

3. Williams, E., Trono-Figueras, R., and Darmofal, D., "Learning Diffusion for Stabilizing Linearized Chaotic Systems," SIAM Annual Meeting, Jul 2024.

4. Williams, E., and Darmofal, D., "Towards a stochastic subgrid-scale model for turbulence," Advances in Computational Mechanics, UT Austin, Oct 2023.

5. Williams, E., Arranz, G., and Lozano-Duran, A., "Wall-Modeled Large-Eddy Simulation of the Lockheed Martin X-59 QueSST," APS Division of Fluid Dynamics, Nov 2022.

6. Williams, E., and Lozano-Duran, A., "Information-theoretic approach for subgrid-scale modeling for high-speed compressible wall turbulence," AIAA AVIATION Forum, Jun 2022.

7. Williams, E., and Lozano-Duran, A., "Error Scaling of Wall-Modeled Large-Eddy Simulation of Compressible Wall Turbulence," APS Division of Fluid Dynamics, Nov 2021.

Teaching & Mentoring

Graduate Mentor – MIT Summer Research Program (MSRP)	2023
Graduate Mentor – MIT Undergraduate Research Opportunities Program (UROP)	2022
Head Engineering Learning Assistant – UIUC Engineering Orientation	2019 – 2021
Course Developer & Instructor – UIUC Aerospace CubeSat Design	2020
Teaching Assistant – UIUC First-Year Engineering Leadership	2020 – 2021
Course Assistant – UIUC Programming for Engineers	2018 – 2021
Lead Tutor – UIUC Center for Academic Resources in Engineering	2019 – 2021

Service & Outreach

President – MIT Graduate Women in Aerospace Engineering	2022
Social Committee Chair – MIT Aerospace Computational Design Lab	2021 – 2022
President – UIUC Women in Aerospace	2019 – 2021